

**File:** 2026-01-14\_Bob-Martinez\_Proteomics-Integration.pdf

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**Project:** Multimodal Neurodegeneration Biomarker Integration

**Section:** CSF and Volumetric Correlation Audit

### 1. Scope and Objective

Objective of this audit is to validate cross-modal consistency between CSF proteomics biomarkers and MRI-derived volumetrics, focusing on CSF Amyloid-beta 42 (A $\beta$ 42) versus hippocampal atrophy. A $\beta$ 42 depletion is a canonical biomarker for amyloid plaque deposition, while hippocampal atrophy represents downstream neurodegeneration. Data sources include CSF proteomics spreadsheet v3.2 and T1-weighted MRI processed using FreeSurfer v7.4.

### 2. MRI Volumetric Processing Summary

T1-weighted MRI preprocessing included bias field correction and skull stripping, followed by segmentation using FreeSurfer recon-all. Bilateral hippocampal volumes were extracted and normalized by intracranial volume. Manual QC was performed on all segmentations. Normalized hippocampal volume was expressed as percent deviation from age-matched healthy controls.

### 3. Statistical Analysis: A $\beta$ 42 vs Hippocampal Atrophy

Pearson and Spearman correlations were computed across 143 subjects with complete data. A strong negative correlation was observed (Pearson  $r = -0.71$ ,  $p < 1e-6$ ; Spearman  $\rho = -0.68$ ). Lower CSF A $\beta$ 42 levels were consistently associated with increased hippocampal atrophy, persisting after correction for age, sex, and scanner site.

### 4. Cross-Modal Consistency Check

Observed trends align with the amyloid cascade hypothesis and prior internal atrophy models. No evidence of scanner-driven artifacts or label leakage was detected. CSF A $\beta$ 42 is behaving as expected relative to structural neurodegeneration.

## 5. MAJOR DATA INTEGRITY ISSUE (BLOCKER)

### 5.1 Tau-Protein Column Completeness

The Tau-protein column in the proteomics dataset is approximately 15% empty, with missingness concentrated in the healthy control group. Affected participant IDs range from AD-102 to AD-118. Missingness is likely missing-at-random at best, potentially site-related.

### 5.2 Impact on Downstream Modeling

This issue blocks Chloe's multimodal fusion model, where Tau is a required input. Proceeding without remediation would bias feature distributions and invalidate cross-validation. Training in PyTorch cannot proceed safely with this level of structured missingness.

### 5.3 Remediation Options (Decision Required)

Option A: K-Nearest Neighbors imputation preserves sample size but introduces synthetic signal and smoothing risk. Option B: Excluding participants AD-102 to AD-118 ensures clean statistics but reduces healthy control representation and increases class imbalance.

## 6. Action Items

Alice: advise on preferred strategy (KNN imputation vs exclusion). Chloe: keep fusion model training paused. Bob: ready to implement either remediation pathway and provide follow-up audit reports.

## 7. Status Summary

CSF A $\beta$ 42 to hippocampal atrophy relationship is validated. Tau-protein missingness remains unresolved and blocks modeling pending decision.